Homework 6: Metals

Paper track. No home lab (the metals kit stays at practice). Calculator needed.

Part A: Identify the metal (2 pts each)

#	Color	Magnet	Density (g/cm^3)	With HCl	Metal + symbol + atomic #
1	reddish-orange	no	8.96	no reaction	
2	silvery	no	2.70	slow start, then steady	
3	gray	**yes**	7.87	slow–moderate, solution goes pale green	
4	dull silver	no	1.74	vigorous, fast	
5	bluish-gray	no	7.14	moderate, steady	

Part B: Density (3 pts each)

1. A metal cube 3.0 cm on a side weighs 72.9 g. Density? Which metal?
2. A piece of metal raises water in a graduated cylinder from 25.0 mL to 34.0 mL and weighs 80.6 g. Density? Which metal?
3. Convert 7.14 g/cm^3 to kg/m^3.
4. A solid iron bar has a volume of 50 cm^3. What is its mass? (Fe = 7.87 g/cm^3)
5. A 2.0-meter cube of aluminum — what is its mass in kg? (Al = 2700 kg/m^3)

Part C: Reactions & oxides (2 pts each)

1. Classify: $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$
2. Classify: $Zn + 2\ HCl \rightarrow ZnCl_2 + H_2$
3. Classify: $2\ Mg + O_2 \rightarrow 2\ MgO$

4. Why does aluminum resist acid at first, but iron just keeps corroding?
5. Steel, brass, and bronze — name the metals in each.

Part D: Scenario (3 pts)

A gray metal fragment at a scene is not attracted to a magnet, has a measured density of 2.71 g/cm^3, and reacts slowly with HCl at first and then steadily. Name the metal. A suspect who works with aircraft parts and a suspect who works with cast-iron cookware are both of interest — which does this implicate?